Cognitive Modeling: Homework Assignment 3
Stochastic Process Models and Bayesian Estimation

February 28, 2026

All answers and solutions to non-programming questions should be submitted to LMS as a legible write-up (either fully digital or a scan). The use of LLMs (e.g., ChatGPT) is explicitly discouraged, unless specified otherwise. All code should be committed to and merged into the main branch of your team's GitHub repository, unless specified otherwise. Your LMS submissions should contain a single ZIP file named according to the pattern:

- CogModel Assignment[#] [TeamMember1Initials] [TeamMember2Initials]

Only one team member should initiate the submission on LMS. A link to the GitHub repository should be clearly visible on the LMS as submission text. :contentReference[oaicite:0]index=0

Problem 1: True-False Questions (5 points)

Mark all statements which are FALSE.

1. The solution of the stochastic integral $\int_0^T \mu \, dW_t$ is $\mu(W_T - W_0)$ and is a random variable itself.
2. **The variance of a Wiener process with scale coefficient $\sigma = 1$ at time t is t^2.**
3. **The standard Drift-Diffusion Model (DDM) assumes that evidence about a dominant alternative accumulates in discrete chunks over time.**
4. **The first passage time distribution has a closed-form probability density function, but its density can still be evaluated only numerically.**
5. **The Euler-Maruyama method can only be used to simulate linear stochastic systems.**
6. **For any Bayesian analysis, the prior will always have a smaller variance than the posterior.**
7. In addition to good statistical practices, experimental validation of cognitive models is crucial for ensuring construct validity.
8. **Markov chain Monte Carlo (MCMC) methods approximate a complex posterior distribution through a simpler, yet analytically tractable, distribution.**

9. For most Bayesian problems, the more data we collect, the less influence does the prior exert on the resulting inferences.

10. The effective sample size (ESS) estimated from MCMC samplers differs from the total number of samples because the samples are not independent.

Problem 2: Diffusion Model Explorations (8 points)

As extensively discussed in class, the drift-diffusion model (DDM) generates two response time (RT) distributions, one for each boundary (i.e., lower and upper boundaries). This exercise asks you to first explore a somewhat counterintuitive question about the basic DDM: What differences between the means of the two RT distributions does the model predict?

To approach this question from a simulation-based perspective, you need to repeatedly solve the forward problem with different parameter configurations and collect the two summary statistics, namely, the two empirical means of the resulting RT distributions. First, choose a suitable configuration of the four parameters and vary only the drift rates within a reasonable range (e.g., $v \in [0.5, 1.5]$) for a total of 25 different drift rates. Make sure that your parameterizations can generate a sufficient number of RTs for both distributions and you don't end up with the process only reaching the upper boundary. Second, for each of your parameter configurations, generate $N = 2000$ synthetic observations and estimate the means of the two distributions. What do you observe regarding the mean difference? Describe and interpret your results. (4 points)

In a similar spirit (keeping all parameters fixed and varying one), explore the effects of each of the parameters on the means and standard deviations of the simulated RT distributions, quantify and describe your results. (4 points)

Code and report are included in the Q2_Model.ipynb.

Problem 3: Prior and Posterior Variance (4 points)

Show that the following identity holds for any given prior and posterior:

$$\mathrm{Var}[\theta] = \mathbb{E}[\mathrm{Var}(\theta \mid y)] + \mathrm{Var}(\mathbb{E}[\theta \mid y])$$

Clarification of terms:

1. $\mathrm{Var}[\theta]$ – Prior variance.

2. $\mathbb{E}[\mathrm{Var}(\theta \mid y)]$ – Expected posterior variance.

3. $\mathrm{Var}(\mathbb{E}[\theta \mid y])$ – Variance of posterior mean.

By definition:

$$\mathrm{Var}[\theta] = \mathbb{E}[\theta^2] - (\mathbb{E}[\theta])^2$$

Applying the law of total expectation:

$$\mathbb{E}[\theta^2] = \mathbb{E}[\mathbb{E}[\theta^2 \mid y]], \quad \mathbb{E}[\theta] = \mathbb{E}[\mathbb{E}[\theta \mid y]]$$

Thus:

$$\mathrm{Var}[\theta] = \mathbb{E}[\mathbb{E}[\theta^2 \mid y]] - (\mathbb{E}[\mathbb{E}[\theta \mid y]])^2$$

From:

$$\mathrm{Var}(\theta \mid y) = \mathbb{E}[\theta^2 \mid y] - (\mathbb{E}[\theta \mid y])^2$$

we can rewrite:

$$\mathbb{E}[\theta^2 \mid y] = \mathrm{Var}(\theta \mid y) + (\mathbb{E}[\theta \mid y])^2$$

Substituting into $\mathrm{Var}[\theta]$:

$$\mathrm{Var}[\theta] = \mathbb{E}[\mathrm{Var}(\theta \mid y)] + \mathbb{E}[(\mathbb{E}[\theta \mid y])^2] - (\mathbb{E}[\mathbb{E}[\theta \mid y]])^2$$

From a previous step:

$$\mathrm{Var}(\mathbb{E}[\theta \mid y]) = \mathbb{E}[(\mathbb{E}[\theta \mid y])^2] - (\mathbb{E}[\mathbb{E}[\theta \mid y]])^2$$

final result:

$$\mathrm{Var}[\theta] = \mathbb{E}[\mathrm{Var}(\theta \mid y)] + \mathrm{Var}(\mathbb{E}[\theta \mid y])$$

Problem 4: Normal-Normal Model (4 points)

This exercise mimics the approach we took for solving the Beta-Binomial Bayesian butter toast example using a Normal-Normal conjugate model with known variance (i.e., Normal prior, Normal likelihood). You would implement the model in Python (just as we did for the butter toast) and explore the effects of different prior configurations, so you don't necessarily need to analytically derive the results.

The prior distribution for the mean is given by:

$$p(\mu) = \mathcal{N}(\mu \mid \mu_0, \sigma_0)$$

The likelihood distribution is given by:

$$p(y \mid \mu, \sigma^2) = \mathcal{N}(y \mid \mu, \sigma^2)$$

Come up with an entertaining use case for the model, simulate a modest number of data points, and compute the posterior given a self-chosen configuration for the prior. Visualize the difference between prior and posterior using histograms. (4 points)

Histogram png included in zip file.

Use Case: The average time a student takes to answer a test question

Problem 5: Simple Bayesian Regression with Stan (6 points)

All problem 5 code is zipped with PDF/located in Github repo.

Model and Simulation Setup

I fit the model

$$\begin{aligned}
\sigma^2 &\sim \text{Inv-Gamma}(1, 1), \\
\alpha &\sim \mathcal{N}(0, 10), \\
\beta &\sim \mathcal{N}(0, 10), \\
y_n &\sim \mathcal{N}(\alpha + \beta x_n, \sigma^2),
\end{aligned}$$

with simulated data where the covariate is standard normal, $x_n \sim \mathcal{N}(0,1)$, and

$$y_n = \alpha + \beta x_n + \varepsilon_n, \quad \varepsilon_n \sim \mathcal{N}(0, \sigma^2).$$

I used values that are easy to interpret on a generic outcome scale:

$$\alpha = 10, \quad \beta = 3, \quad \sigma = 3$$

Numerical Posterior Results

Table 1 summarizes all of the posterior data from the runs.

Parameter	N	Posterior mean	Posterior SD	97.5% CrI
α	100	10.086	0.289	[9.525, 10.657]
β	100	2.691	0.307	[2.087, 3.292]
σ	100	2.919	0.206	[2.553, 3.354]
α	1000	9.993	0.095	[9.809, 10.178]
β	1000	3.174	0.094	[2.991, 3.355]
σ	1000	3.012	0.067	[2.886, 3.147]

Table 1: Posterior summaries for simulated truth $(\alpha, \beta, \sigma) = (10, 3, 3)$.

We need to check the convergence and Efficiency diagnostics to make sure that the date is suitable.

$\hat{R}$ and ESS Values

- $N = 100$: max $\hat{R} = 1.00073$, min bulk ESS $= 3764$.
- $N = 1000$: max $\hat{R} = 1.00108$, min bulk ESS $= 3987$.

No divergent transitions or other warnings. The posteriors are computationally faithful and suitable for inference.

Conclusions from the Data

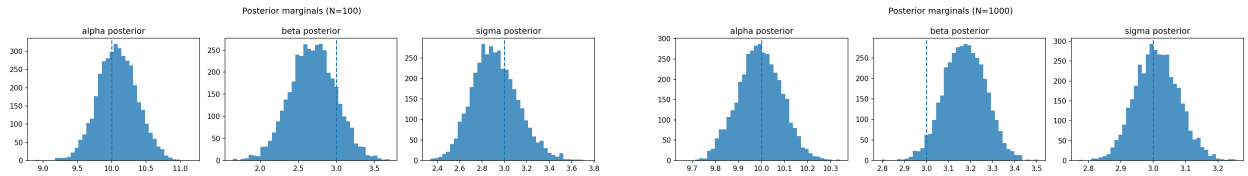
How accurate are posterior means and how much uncertainty is left? From our original parameters $(\alpha, \beta, \sigma) = (10, 3, 3)$, we can see that the posterior means are relatively close to the original values. Out of all of our values, β seems to be the furthest away from our original value. For N = 100, there still seems to be a degree of uncertainty.

What happens with ten times more observations? After 10 times the amount of operations, we can see that we have refined are values for α β σ to be a lot closer to our original. We can also see that our uncertainty (represented by the Posterior SD and 95% CrI) is also a lot smaller, meaning that we are a lot more confident in our output.

Figures

Problem 6: Estimating the Drift-Diffusion Model (8 points)

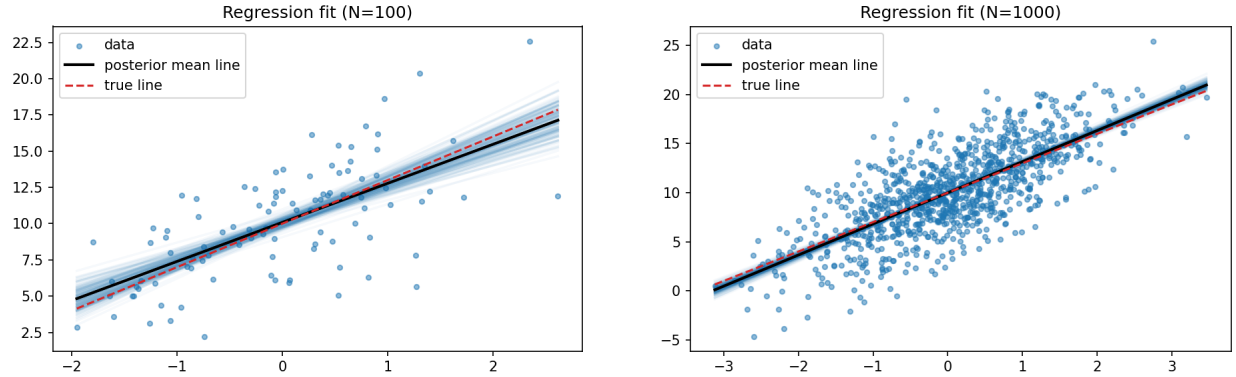
All problem 6 code is zipped with PDF/located in Github repo.



(a) $N = 100$: posterior marginals for α, β, σ.

(b) $N = 1000$: posterior marginals.

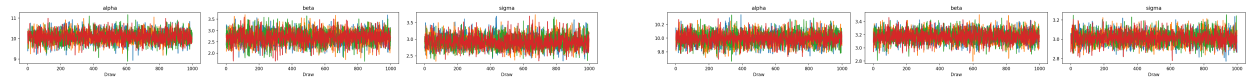
Figure 1: Posterior distributions (dashed lines: true $\alpha = 10$, $\beta = 3$, $\sigma = 3$).



(a) $N = 100$: data, posterior mean regression line, and true line.

(b) $N = 1000$: same.

Figure 2: Regression fit.



(a) $N = 100$ chains.

(b) $N = 1000$ chains.

Figure 3: Trace plots for convergence diagnostics.

Which parameter reflects task difficulty?

In this problem, the task difficulty (quality and rate of information accumulation) is best reflected by the drift rate v. Lower drift means noisier evidence. With this established, I allowed drift to vary by condition, while keeping all other parameters shared. After running the program, I then compared the two drift rates:

Posterior mean drift, averaged across operators:

- Condition 1: $\bar{v} \approx 0.58$
- Condition 2: $\bar{v} \approx 3.47$

Which condition is high-interference?

Comparing the 2 conditions, we can see that the drift rate from condition 2 is much higher, meaning that the processing was much easier.

We can then conclude that condition 1 is the high-interference (difficult) field while condition 2 is the low-interference field.

$\hat{R}$ and ESS Values

We need to check the convergence and Efficiency diagnostics to make sure that the results are suitable.

- Max $\hat{R} = 1.006$, minimum bulk ESS ≈ 1659

No divergent transitions or other warnings. The posteriors are computationally faithful and suitable for inference.